FLAT PULLBACK OF THE DIVISOR OF A MEROMORPHIC SECTION IS THE MULTIPLICITY WEIGHTED SUM OVER THE COMPONENTS OF THE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a reduction to the trivialising torsor followed by component and multiplicity bookkeeping and the identification of the sections with rational functions on the torsor

Editorial note. The label lemma-prepare-flat-pullback-cap-c1 names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The divisor of a nonzero meromorphic section of an invertible sheaf and its flat pullback, together with the multiplicity of an irreducible component in a cycle
- The construction of a torsor trivialising an invertible sheaf, the identification of the divisor of a meromorphic section with the divisor of the corresponding rational function on that torsor, and the behaviour of the divisors under the projection
- The composition of flat pullbacks, and the rational function case of the formula for flat pullback of a principal divisor with multiplicities

Source reference. Original reference 0EQG: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 2385–2385. Prerequisite pointer invoked at proof line(s) 2757; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EQH: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 2477–2477. Prerequisite pointer invoked at proof line(s) 2778; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02RD: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1347–1347. Prerequisite pointer invoked at proof line(s) 2779; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0EPH: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 2039–2039. Prerequisite pointer invoked at proof line(s) 2753; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 2736; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 02QL let $X, Y/B$ be good. Let $f : X \to Y$ be a flat morphism of relative dimension r. Let $\mathcal{L}$ be an invertible sheaf on Y. Assume Y is integral and $n = \dim_\delta(Y)$. Let s be a nonzero meromorphic section of $\mathcal{L}$. Then we have*

$$f^* div_{\mathcal{L}}(s) = \sum n_i div_{f^*\mathcal{L}|_{X_i}}(s_i)$$

in $Z_{n+r-1}(X)$. Here the sum is over the irreducible components $X_i \subset X$ of δ-dimension $n + r$, the section $s_i = f|^*_{X_i}(s)$ is the pullback of s, and $n_i = m_{X_i,X}$ is the multiplicity of X_i in X.

Proof. Using sleight of hand we will deduce this from Lemma 0EPH. (An alternative is to redo the proof of that lemma in the setting of meromorphic sections of invertible modules.) Namely, let $q : T \to Y$ be the morphism of Lemma 0EQG constructed using $\mathcal{L}$ on Y. We will use all the properties of T stated in this lemma. Consider the fibre product diagram

$$\begin{array}{ccc} T' & \xrightarrow{q'} & X \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle f} \\ T & \xrightarrow{q} & Y \end{array}$$

Then $q' : T' \to X$ is the morphism constructed using $f^*\mathcal{L}$ on X. Then it suffices to prove

$$(q')^* f^* \mathrm{div}_{\mathcal{L}}(s) = \sum n_i (q')^* \mathrm{div}_{f^*\mathcal{L}|_{X_i}}(s_i)$$

Observe that $T'_i = q^{-1}(X_i)$ are the irreducible components of T' and that n_i is the multiplicity of T'_i in T'. The left hand side is equal to

$$h^* q^* \mathrm{div}_{\mathcal{L}}(s) = h^* \mathrm{div}_T(q^*(s))$$

by Lemma 0EQH (and Lemma 02RD). On the other hand, denoting $q'_i : T'_i \to X_i$ the restriction of q' we find that Lemma 0EQH also tells us the right hand side is equal to

$$\sum n_i \mathrm{div}_{T_i}((q'_i)^*(s_i))$$

In these two formulas the expressions $q^*(s)$ and $(q'_i)^*(s_i)$ represent the rational functions corresponding to the pulled back meromorphic sections of $q^*\mathcal{L}$ and $(q'_i)^* f^* \mathcal{L}|_{X_i}$ via the isomorphism $\alpha : q^*\mathcal{L} \to \mathcal{O}_T$ and its pullbacks to spaces over T. With this convention it is clear that $(q'_i)^*(s_i)$ is the composition of the rational function $q^*(s)$ on T and the morphism $h|_{T'_i} : T'_i \to T$. Thus Lemma 0EPH exactly says that

$$h^* \mathrm{div}_T(q^*(s)) = \sum n_i \mathrm{div}_{T_i}((q'_i)^*(s_i))$$

as desired. □